# Rahul Nihalani


AI/ML Engineer | Generative AI | NLP | MLOps

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## Summary


Final-year M.Tech Artificial Intelligence student and AI/ML Engineer specializing in Generative AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), NLP, and MLOps. Experienced in building production-ready AI applications using FastAPI, React, PostgreSQL, Redis, Qdrant, Docker, AWS, and CI/CD pipelines, with published research in NLP.


## Experience

**Grammatical Error Detection (GED) (arXiv)** — July 2023 – June 2024

*NLP Intern, Sitare University* — *Remote, India*

- Designed an 8-step data cleaning pipeline for the Lang-8 dataset, reducing volume from 2.37M to 200K sentences and improving GED model performance by 15%.
- Fine-tuned BERT-base, BERT-large, RoBERTa-base, and RoBERTa-large models for GED; BERT-base achieved 90.53% accuracy and 0.91 F1-score, outperforming larger transformer variants.
- Benchmarked transformer and LLM-based approaches including GPT-4 and Llama-3-70B-Instruct on 500 evaluation samples, achieving up to 0.99 F1-score and a 9% accuracy improvement.

## Projects

**Enterprise AI Knowledge Assistant (EAIKA) (GitHub) (App)** — June 2026 – Present

*Full-Stack GenAI Rag Platform* — *FastAPI, React, PostgreSQL, Redis, Qdrant, Docker*

- Engineered a full-stack RAG platform using FastAPI, React, PostgreSQL, Redis, and Qdrant, supporting document ingestion, semantic search, conversational AI, authentication, analytics, and real-time WebSocket communication.
- Implemented hybrid retrieval using BM25, dense vector search, reranking, semantic chunking, and agent-based workflows, improving retrieval relevance and reducing hallucinations in LLM-generated responses.
- Built production-grade infrastructure with Docker, GitHub Actions CI/CD, Celery workers, Redis caching, Prometheus monitoring, rate limiting, and Kubernetes-ready deployment architecture.

**Indian Sign Language Recognition using LSTM Model (GitHub)** — June 2022 – October 2022

*Deep Learning Project* — *Python, TensorFlow, OpenCV*

- Led a team of five to develop a real-time Indian Sign Language recognition system using TensorFlow and OpenCV, trained on a custom dataset of 200K gesture images from eight participants.
- Developed a 14-layer LSTM architecture achieving 94.28% training accuracy and 91.50% evaluation accuracy, outperforming CNN-based baselines for sequential gesture classification.
- Curated and annotated a custom digit recognition dataset (0–9), optimizing model training with Adam and categorical cross-entropy, resulting in an 8% improvement in classification precision.

## Education

**Vellore Institute of Technology** — Bhopal, India

*Integrated M.Tech in Artificial Intelligence | GPA: 9.16/10* — *September 2021 – Expected October 2026*

## Technical Skills

**Languages**: Python, Java, SQL, JavaScript, TypeScript

**AI/ML**: PyTorch, TensorFlow, Scikit-Learn, Transformers, LangChain, LlamaIndex, NLP, Computer Vision, LLMs, Generative AI

**LLM Engineering**: RAG, Agentic AI, Prompt Engineering, Fine-Tuning, LLM Evaluation, Vector Databases, Semantic Search

**Web & Backend**: React, HTML, CSS, Tailwind CSS, Flask, FastAPI, REST APIs, WebSockets

**Databases & Retrieval**: PostgreSQL, MongoDB, Redis, Qdrant, FAISS, Semantic Search, Hybrid Retrieval

**Cloud & MLOps**: AWS (EC2, ECR, S3, IAM, RDS, Lambda, SageMaker), Docker, GitHub Actions, MLflow, DVC, Airflow, CI/CD, Grafana, Prometheus

## Publications

- Co-authored with Prof. Kushal Shah (Ex-IIT Delhi) and published research on Grammatical Error Detection (GED) on arXiv, evaluating transformer and LLM-based approaches for automated grammar assessment.

## Certifications

- Advanced Large Language Models (LLMs), Self Shiksha & Pariksha.
- Applied Machine Learning in Python (95.28%), Coursera.